```python
import numpy as np
import pandas as pd
from sklearn.datasets import make_classification
from sklearn.ensemble import RandomForestClassifier
from sklearn.feature_selection import SelectFromModel
from sklearn.model_selection import train_test_split

# 1. Generate a dummy classification dataset for demonstration
X, y = make_classification(
    n_samples=1000, n_features=20, n_informative=5, n_redundant=5, random_state=42
)
feature_names = [f"feature_{i}" for i in range(X.shape[1])]
X_df = pd.DataFrame(X, columns=feature_names)

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X_df, y, test_size=0.3, random_state=42
)

# 2. Fit a Random Forest to calculate feature importances
clf = RandomForestClassifier(n_estimators=100, random_state=42)
clf.fit(X_train, y_train)

# 3. Use SelectFromModel to automatically drop low-importance features
# 'median' threshold will drop the 50% least important features
selector = SelectFromModel(clf, threshold="median", prefit=True)

# 4. Transform data to keep only useful features
X_train_selected = selector.transform(X_train)
X_test_selected = selector.transform(X_test)

# Get the names of the selected features for production deployment
selected_features = X_df.columns[selector.get_support()]

# Print original and reduced feature details
print(f"Original number of features: {X_train.shape[1]}")
print(f"Original features considered: {list(X_df.columns)}")
print("-" * 50)
print(f"Reduced number of features: {X_train_selected.shape[1]}")
print(f"Selected features for production: {list(selected_features)}")
```

```
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2732: UserWarning: X has feature names, but SelectFromModel was fitted without feature names
  warnings.warn(
/usr/local/lib/python3.13/dist-packages/sklearn/utils/validation.py:2732: UserWarning: X has feature names, but SelectFromModel was fitted without feature names
  warnings.warn(
Original number of features: 20
Original features considered: ['feature_0', 'feature_1', 'feature_2', 'feature_3', 'feature_4', 'feature_5', 'feature_6', 'feature_7', 'feature_8', 'feature_9', 'feature_10', 'feature_11', 'feature_12', 'feature_13', 'feature_14', 'feature_15', 'featur
--------------------------------------------------
Reduced number of features: 10
Selected features for production: ['feature_0', 'feature_5', 'feature_8', 'feature_9', 'feature_10', 'feature_11', 'feature_13', 'feature_14', 'feature_15', 'feature_18']
```